Florida Estuarine Restoration Database

Database Development Protocol

Version 1.0.0

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Developed by: Gulf Ecosystem Initiative Restoration and Fisheries Working Group
Project partners: National Center for Ecological Analysis and Synthesis (NCEAS)

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1. Introduction and objectives

This document outlines the methods used to develop and maintain the Florida Estuarine Restoration Database. The database was developed by the National Center for Ecological Analysis and Synthesis (NCEAS) Gulf Ecosystem Initiative Restoration and Fisheries Working Group to support the group's objectives of evaluating the impacts of large-scale estuarine habitat restoration on fisheries in the Gulf of Mexico. Its structure and project inclusion criteria were informed by those developed for the working group's associated systematic literature review.

The database provides a comprehensive, spatially consistent record of estuarine habitat restoration projects in Florida's estuaries and its associated watersheds. It is intended to support quantitative analyses of restoration activity, including assessments of the spatial and temporal distribution of projects, restoration investments, and habitat coverage.

This protocol documents the procedures used to develop version 1.0.0 of the Florida Estuarine Restoration Database. It provides a reference for understanding the database structure and methods used to compile, standardize, and integrate project records. Future versions of the protocol may be revised to reflect changes in data sources, database structure, or processing methods.

Variable definitions, units, and other field-level information are documented separately in the accompanying metadata definition document.

All data processing, standardization, and analysis were conducted in R Programming[1] (version 4.4.1). Geographic screening and spatial processing were conducted in QGIS[2] (version 3.34.1).

2. Methods

2.1 General framework

The Florida Estuarine Restoration Database was developed through a systematic process involving source dataset identification, compilation, standardization, screening, integration, and deduplication. Relevant datasets were identified and obtained through direct downloads from publicly available platforms or through data requests to source organizations and agencies.

Source datasets were converted to a standardized structure, with variables recoded and reformatted as necessary to align with the variable definitions established for the database. Project records were screened against predefined inclusion criteria, and source datasets were assessed for missing or incomplete information. Where feasible, source records were supplemented using supporting documentation or additional calculations to improve data completeness.

Standardized source datasets were integrated into a centralized database. Additional processing was performed to standardize terminology and values within selected variables. Because source datasets varied in their reporting objectives, database structures, and levels of project detail, a systematic deduplication procedure was developed to identify, evaluate, and reconcile duplicate project records within and across source datasets.

2.2 Source dataset compilation

Source datasets containing information on restoration projects in Florida were identified through the expertise and knowledge of working group members. Eleven source datasets were compiled, including publicly accessible databases, agency-maintained project platforms, and internal agency datasets requiring direct requests for access (Table 1).

Where available, datasets were downloaded directly from public platforms using existing queries to retrieve Florida-specific project records. In some cases, downloaded files contained only URLs linking to individual project webpages rather than the associated project information. For these sources, data requests were submitted to the relevant agency to obtain project-specific information in spreadsheet format. When requesting data directly, priority was given to obtaining detailed information about individual habitat restoration projects.

Some sources required manual compilation from primary source documentation, including PDF reports. For these sources, only projects that met the inclusion criteria or were clearly focused on restoration were compiled; other projects were excluded. Source metadata were retained when

available for datasets obtained through direct downloads or agency requests. Standardized metadata were not consistently available for manually compiled sources.

Datasets were classified as primary or secondary sources according to how the project information was obtained. Primary sources consisted of information obtained directly from the responsible agency or public data platform. One exception was a static dataset compiled by a former NCEAS coastal restoration working group, which aggregated restoration project information from other primary sources and was therefore treated as a secondary source. This dataset was used as a supplementary reference to cross-check records and identify potentially relevant restoration projects not represented in the primary datasets. Records from the secondary source were evaluated alongside primary-source records during database compilation; however, the secondary dataset was not used as a substitute for information available directly from primary sources.

Table 1. Original data sources used to compile the Florida Estuarine Restoration Database, including data source codes, acquisition methods, and most recent access dates. Data source codes were assigned to each original dataset and retained within the database to document the origin of project data.

Data source code	**Description**	**Data acquisition**	**Most recent access date *(version)***	**Reference**
DIVER	National Oceanic and Atmospheric Administration's (NOAA) Data Integration Visualization Exploration and Reporting (DIVER) Explorer for Deepwater Horizon restoration projects	All project data were initially downloaded directly from the public platform (query: *Implementation Group = Florida Trustee Implementation Group*). The CSV export did not include all the project specific information provided on the platform, and so a direct data request was made to the agency to obtain the project specific information as a CSV and avoid manual data extraction from the platform.	02/04/2026	3
DWH	Deepwater Horizon Project Tracker	All project data were directly downloaded from the public platform Direct download (query: *State Label = Florida*).	12/11/2025	4
FDEP	Florida Department of Environmental Protection's (FDEP) Environmental Resource Permitting (ERP) program	Direct data request to agency for all environmental resource permits in Florida.	05/19/2025	5
FWC_MESS	Florida Fish and Wildlife Conservation Commission's (FWC), Division of Habitat and Species Conservation, Aquatic Habitat Conservation and Restoration Section, Marine/Estuarine Subsection (MESS!)	Manual extraction from project PDFs via public platform (query: *Project_type* contains *'Restoration'*).	03/01/2026	6
FWC_REEFS	Florida Fish and Wildlife Conservation Commission's (FWC) Artificial Reef Program	All project data were initially downloaded directly from the public platform. Agency staff offered to conduct additional calculations and provide a supplemental dataset containing enhanced project specific information not available in the public download (see section 4.4). Spatial boundaries were provided to the agency in advance of the calculations to minimize the workload on agency staff.	06/23/2026	7,8

NACC	National Aquatic Connectivity Collaborative (NACC) Barrier Inventory & Prioritization Tool	All project data were downloaded directly from the public platform Direct download (query: *Florida*)	02/04/2026 (*v4.0.0)*	9
NEP	National Estuary Program (NEP)	2010-present: Project data are publicly available through the National Estuary Program Dashboard Pre-2010: Project data were accessed from NEPORT database by NEP staff (query: *State_Code = FL*); pre-2006 project data were collected differently and compiled through various NEP specific projects.	01/06/2026	10
NFWF_NCRF andECRF	National Fish and Wildlife Foundation's (NFWF) Coastal Resilience Dashboard (funding through National Coastal Resilience Fund or Emergency Coastal Resilience Fund)	All project data were initially downloaded directly from the public platform (query: *State = Florida*). The CSV export included limited project specific information, and so additional project specific information was obtained through direct request and manual data extraction of NFWF reports (see section 4.4).	08/11/2026	11
NFWF_GEBF	National Fish and Wildlife Foundation's (NFWF) (funding through Gulf Environmental Benefit Fund)	Manual extraction from NFWF website project PDFs for all estuarine related projects in Florida. Additional project specific information was obtained through direct request (see section 4.4)).	07/01/2026	12
NOAA	National Oceanic and Atmospheric Administration (NOAA) Restoration Center's Restoration Atlas for habitat restoration projects	All project data were initially downloaded directly from the public platform (query: *State = FL*). The CSV export did not include all the project specific information provided on the platform, and so a direct data request was made to the agency to obtain the project specific information as a CSV and avoid manual data extraction from the platform.	03/19/2026	13
SNAPP	The Science for Nature and People Partnership (SNAPP) - National Center for Ecological Analysis and Synthesis (NCEAS) joint working group: Aligning coastal restoration with ecological and societal needs	Direct data request of previously compiled CSV of coastal wetland restoration projects in Florida.	04/21/2025	14

2.3 Project screening criteria

Potential projects identified in each source dataset were assessed against predefined project eligibility criteria and geographic boundaries before inclusion in the database. During initial compilation and screening, projects were excluded when the available information clearly indicated that they did not meet the eligibility criteria or were located outside the defined geographic extent.

A final spatial screening was conducted after database cleaning and deduplication to verify the geographic eligibility of the resulting project records. This additional screening accounted for the possibility that merging duplicate records from different sources could introduce new coordinates or project descriptions that affected the assessment of geographic eligibility.

2.3.1 Project inclusion and exclusion criteria

Projects were included when they involved estuarine habitat restoration in Florida estuaries and associated watersheds. Eligible activities primarily consisted of habitat restoration but also included activities categorized by their source datasets as habitat enhancement, habitat creation, or conservation.

Projects were excluded when they focused exclusively on restoration program development or scientific research without associated on-the-ground implementation. Projects focused exclusively on stormwater runoff management were also excluded.

The database was designed to document restoration implementation where on-the-ground outcomes were planned, regardless of whether projects ultimately achieved their intended ecological performance targets. Therefore, projects were retained even when restoration efforts failed to meet their intended outcomes. For example, projects involving repeated replanting efforts were included even when the target percentage of spatial coverage was not achieved within the designated funding period.

Projects funded solely for the planning or design of future restoration activities were excluded. Projects were also excluded when funding had been terminated before any restoration activity occurred, or when projects had been reported as inactive for more than five years because of infeasibility or insufficient financial support, with no alternative plan reported. Projects for which permit applications had been denied or formally withdrawn were excluded.

No explicit temporal boundaries were applied. Projects were eligible regardless of their reported completion year or source or amount of funding, provided they met the other inclusion criteria.

2.3.2 Geographic boundaries and spatial screening

Geographic eligibility was defined to capture restoration activities occurring within Florida estuaries and their associated watersheds. Each project was assessed using a single reported project location. The geographic boundaries were established to encompass both coastal estuarine habitats and inland areas where restoration activities could contribute to downstream aquatic systems.

For this database, estuarine habitats included subtidal and intertidal environments such as seagrass, oyster, salt marsh, mangrove, hard-bottom, tidal river and creek, coastal wetland, and tidal flat habitats. Because estuaries are transitional coastal systems where freshwater and saline water mix, their seaward boundaries are not consistently defined across available spatial datasets. The seaward extent of the eligible geographic area was therefore delineated using the Florida Department of Environmental Protection's Estuary Nutrient Regions[15], which provide predefined coastal management boundaries along the Florida coastline. Regions characterized primarily by marine environments with few estuarine features were excluded from the eligible area. These included the Oceanside, Marquesas, Outer Gulf Shelf, Middle Gulf Shelf, Collier Inshore, Lower Keys, Middle Keys, and Upper Keys regions identified in the Estuary Nutrient Regions shapefile.

To include upstream estuarine restoration activities, the landward extent was defined using the spatial union of Florida's coastal county boundaries[16] and intersecting Hydrologic Unit Code 10 (HUC10) watershed boundaries[17]. Coastal counties captured the broader coastal areas associated with Florida estuaries, while HUC10 watersheds extended the eligible area farther inland. These spatial features were compiled into a geospatial shapefile for geographic screening.

During initial screening, projects were excluded when available project information clearly indicated that their locations fell outside the defined geographic boundaries. Following database cleaning and deduplication, the resulting project records underwent a final spatial screening in QGIS. This step was necessary because merging duplicate records across source datasets could introduce additional coordinates or project description information relevant to geographic eligibility.

Records were excluded during final screening if their reported locations fell outside the defined geographic boundaries or if they lacked coordinates and sufficient spatial information to determine whether they met the geographic eligibility criteria.

2.4 Database standardization and processing

2.4.1 Database structure and project identification

The Florida Estuarine Restoration Database distinguishes between projects and individual

records in order to account for differences in how restoration actions were reported across the different contributing data sources. Information within source datasets, and ultimately the integrated database, are organized into variables containing either character (terms) or numerical (metric) values, with each variable represented as a column.

A *project* represents a distinct restoration effort identified primarily by project name. Start year, completion year, project description, and location were used, where necessary, to distinguish projects with similar names occurring in the same area or during different time periods. Depending on how restoration activities and quantitative metrics were reported by the original source, a project could be represented by one or multiple records in the database.

A *record* represents an individual entry from a source dataset. The original project identifier assigned by the source was retained in the `Original_Project_ID` variable. The number and structure of records associated with a project reflected the reporting resolution of the original source. For example, some sources reported restoration acreage separately by habitat type or location, resulting in multiple records for a single project.

When multiple records represented distinct components of the same restoration project, they were assigned a common `Project_Group_ID`. This identifier linked related records while preserving the original reporting detail. Duplicate records describing the same restoration project across multiple sources were evaluated during deduplication (section 2.6), with a `Merged_Record_ID` used to document the original project identifiers and source datasets associated with each merged record. The `Master_Project_ID` served as the primary project-level identifier for the resulting restoration project.

2.4.2 Variable harmonization

Eleven source datasets containing potentially relevant restoration information were compiled and harmonized into a standardized database structure. Harmonization involved mapping source variables to standardized database variables, recoding values, and transforming data where necessary to ensure consistency across sources.

Source-specific variable mappings, calculation methods, assumptions, and associated limitations were documented in the data harmonization and standardization crosswalk (Table A1). The crosswalk focused on variables requiring additional processing and did not document every variable when its meaning and interpretation were clear. Procedures for variables requiring standardized categorical classification frameworks are described in the following section.

2.4.3 Standardization of categorical variables

The `Habitat_Type` and `Restoration_Technique` variables represented the target habitat restored and the method of restoration, respectively. Values were extracted from structured

source fields when available and mapped to standardized categories to account for differences in terminology and specificity among datasets (Table 2).

When source classifications were broad or ambiguous, project descriptions were reviewed to determine whether a standardized category could be assigned. An ambiguous source term was mapped to a standardized category only when the associated projects were consistently represented by that category. For example, projects reported under the original `Restoration_Technique` term "species enhancement" were reviewed individually. Because these projects consistently represented shellfish restoration activities, the term was mapped to the standardized category "aquaculture (shellfish)."

For selected projects with broad or vague classifications that could not be mapped directly, project descriptions were manually reviewed and the `Habitat_Type` and `Restoration_Technique` values were adjusted when the available information supported reassignment. Records that underwent this manual recategorization were flagged using a binary indicator (`Manual_Recode`).

When structured entries for either variable were missing, categories were inferred from project descriptions only when the relevant information was explicitly stated. Values were left unassigned (NA) when descriptions were ambiguous or insufficient to support a category assignment, avoiding subjective interpretation or unsupported assumptions.

The `Main_Funding_Source` and `Lead_Implementer` variables retained the information reported in the original source datasets and were not independently interpreted or modified. Two additional variables, `Main_Funding_Source_Category` and `Lead_Implementer_Category`, were created to group reported values into standardized categories (Table 3). Funding categories were assigned based on the funding entity, program, or mechanism reported in `Main_Funding_Source`. Implementer categories were assigned based on the organization or agency reported in `Lead_Implementer` as responsible for project implementation.

The same set of standardized categories was applied to both variables, but assignments were made independently. This allowed funding and implementation to be represented separately for a project; for example, a project could be categorized as state-funded and locally implemented. Funding classifications were based on the primary funding source identified in the source data and were not traced beyond that source. Consequently, a project categorized as state-funded could include federal funds passed through a state agency.

Categorization was performed manually and reviewed by multiple working group members as part of quality control.

Table 2. Categorical groupings of input terms for variables `Habitat_Type` and `Restoration_Technique`. Ambiguous terms were only assigned to categories following manual review of project descriptions for all projects originally classified under the given term.

Variable	**Category**	**Term(s)**
Habitat Type	Coastal upland	dune, barrier island, island, beach/dune
Habitat Type	Coastal wetland	tidal wetland(s), coastal wetland
Habitat Type	Estuarine	estuarine, oyster/corals/sav. *Includes projects where the habitat type was originally defined as 'estuarine', and could not be re-categorized into a more specific habitat type category.*
Habitat Type	Estuarine shoreline	beach, estuarine shoreline, other - beach, rocky shoreline
Habitat Type	Estuarine with freshwater and/or upland	*Includes projects where the habitat type was originally defined as either `mix (estuarine with freshwater)`, `mix (estuarine and upland)`, or `mix (estuarine, freshwater and upland)`, and could not be re-categorized into a more specific habitat type category.*
Habitat Type	Freshwater wetland	forested wetland, shrub swamp (non-mangrove), lake/pond, lake, pond, floodplain, freshwater marsh, freshwater wetland, wetland
Habitat Type	Hard bottom	hard bottom, coral reef, artificial reef
Habitat Type	Mangroves	mangrove(s)
Habitat Type	Mud/sand	softbottom mud/sand, soft bottom/sand, soft bottom/mud, mud/sand
Habitat Type	Oysters	oyster reef/shell bottom, oysters, oyster reef
Habitat Type	Salt marsh	salt marsh, marsh, salt barren, salt tern
Habitat Type	Submerged aquatic vegetation	sav (submerged aquatic vegetation), submerged aquatic vegetation (sav), sav, seagrass, submerged aquatic vegetation
Habitat Type	Shell bottom	aquaculture (shellfish), shell bottom, bivalve, scallops
Habitat Type	Tidal tributary/creek/river	riparian, in-stream, riparian zone (non-wetland), fish passage, river, creek, stream, tidal tributary/creek/river
Habitat Type	Upland	upland(s), grassland, field/meadow, forest/woodland, agriculture/ranch land, scrub/scrubland, pine flatwoods

Habitat Type	Upland and freshwater	*Projects where the habitat type was originally defined as `mix (freshwater and upland)`, and could not be re-categorized into a more specific habitat type category.*
Habitat Type	Water column	estuarine water column, water column, freshwater body
Habitat Type	Other	*Projects where the habitat type was originally defined as either `other` and could not be re-categorized into a more specific habitat type category, or where habitat type could not be categorized into a more specific standard habitat type.*
Habitat Type	Unknown	*Projects where the habitat type was not originally defined.*
Restoration Technique	aquaculture (shellfish)	shellfish seeding, oyster gardening, aquaculture (shellfish), species enhancement, species reintroduction (non-plant)
Restoration Technique	conservation/ preservation	land acquisition, easements, conservation/preservation
Restoration Technique	debris removal	debris removal
Restoration Technique	education/recreation	signage, education and outreach, construction for recreational use, research/recommendations/surveys, education/recreation
Restoration Technique	erosion control	rubble stabilization, breakwater, beach nourishment, beach re-nourishment, erosion control, stream bank stabilization
Restoration Technique	hydrologic restoration	berm/dike removal, dam modification/removal, fish passage, storm water/runoff controls, stormwater/runoff controls (only choose for natural habitat), culvert removal, weir removal, weir construction, culvert modification (including replacement), culvert modification (including installation and replacement), aquatic connectivity, berm/dike modification (including replacement), tide gate modification (including replacement), green stormwater infrastructure, floodplain restoration, bulkhead removal, bridge replacement, causeway removal, aquatic connectivity, ditch removal, hydrologic restoration, culvert modification, tidal flow enhancement/improvement, canal water conveyance improvements
Restoration Technique	living shoreline	shoreline stabilization/enhancement, shoreline protection structures, living shoreline

Restoration Technique	planting	vegetation buffer, planting, rain garden creation, or plugging, vegetation planting, revegetation/revegetated, native planting
Restoration Technique	policy instrument	closed areas, boating speed zones, enforcement
Restoration Technique	prescribed burn	prescribed burn, other - prescribed fire, other - prescribed burn, fire management
Restoration Technique	process-based physical restoration	process-based physical restoration, fencing/netting, filling, placement of fill/dredge material, placement of dredged material, contaminant removal/remediation, beach dune barrier island restoration, substrate modification, sand dune restoration, fill removal, seagrass scar filling, regrading, clearing, excavation, wetland creation, bird habitat enhancement, coastal forest restoration, marsh restoration, rehabilitation/creation
Restoration Technique	reef construction	reef construction - artificial materials, reef construction - natural materials, artificial reef deployment, artificial fish habitat reef construction, oyster reef construction, coral reef construction, coral reattachment, coral nursery, reef construction, oyster clutch placement, oyster reef restoration, coral reef restoration, reef creation
Restoration Technique	species control	invasives removal: vegetation, invasives control/removal - vegetation, invasives removal: fauna, invasives control/removal - fauna, invasives control, invasive controls, invasive plant removal, other - invasive species removal, non-native plant removal/control, exotic species removal/control, invasive species maintenance/management
Restoration Technique	stream modification	stream flow modification (including stream pool construction), stream channel rehabilitation, stream channel reconnection/creation, stream restoration, stream modification, stream flow modification
Restoration Technique	other	fish hatchery construction, native plant nursery construction, grey infrastructure

Table 3. Categorical groupings used to standardize the original terms documented in the `Main_Funding_Source` and `Lead_Implementer` variables, including category definitions and representative source inputs.

Category	Definition	Example source inputs
NRDA Settlement	Funding provided through the Natural Resource Damage Assessment (NRDA) settlement funds.	Projects administered through Deepwater Horizon trustees; Florida Trustee Implementation Group
Federal	Funding or implementation administered directly from U.S. federal agencies, federally authorized programs, or national-level federal initiatives, excluding projects funded through Natural Resource Damage Assessment settlements.	United States Fish and Wildlife Service, Environmental Protection Agency, United States Army Corps, Department of Agriculture, Air Force, Department of Defense, National Oceanographic and Atmospheric Administration, National Fish and Wildlife Foundation, National Academies of Sciences Engineering and Medicine Gulf Research Program, Coastal Zone Management Program
State	Funding or implementation administered by state agencies, statewide programs, or state-authorized revenue sources.	Florida Fish and Wildlife Conservation Commission, State of Florida, Florida Forever, Florida Department of Environmental Protection, state Wildlife Grants, Save Our Rivers, PECO, Clean Water Management Trust Fund, Coastal Partnership Initiative (CPI) Grant, state-authorized ad valorem programs, specialty license plate programs
Regional	Funding or implementation administered through multi-jurisdictional regional entities, watershed authorities, or regional partnerships.	Florida Water Management Districts, National Estuary Programs, regional conservation partnerships
Local	Funding or implementation administered through county, municipal, or other local government entities and locally approved funding sources.	Counties, cities, towns, public K-12 schools, local ad valorem programs, voter-approved tax funded initiative programs (e.g., Save Our Indian River Lagoon, Penny for Pinellas), stormwater utilities fees, Board Of County Commissioners, Tourism Development Council, municipal nature preserves, Capital Improvement Program
Nonprofit / NGO	Funding or implementation provided by nonprofit organizations, charitable foundations, or independent tax-exempt entities.	Conservation foundations, land trusts, 501(c)(3) organizations, community foundations, private nonprofit schools, Rotary Clubs, independent nonprofit research organizations

Private	Funding or implementation administered from for-profit entities, private individuals, or non-charitable private organizations.	Private companies that are not 501(c)(3), condominium associations, environmental consulting firms, homeowner, individual donors, private contributions
Academic	Funding or implementation administered by higher educational institutions or affiliated academic research entities.	Universities, colleges, academic research centers
Mixed-Partner	Funding or implementation administered by multiple sources across more than one category where contributions are either equally shared or not clearly attributed. If a primary funding source is clearly identified (>50%), classification is based on that.	Jointly funded federal-state-local projects, multi-organizations partnerships without a designated lead contributor
Other	Non-monitary or unconventional forms of contributions not classified by the categories above.	Volunteer labor, donated services, services in lieu of rent
Unknown	Funding source or implementer is not identified or cannot be determined from available information.	NA, unknown

2.4.4 Standardization of quantitative variables

Quantitative restoration metrics, including `Acres_Restored`, `Miles_Restored`, `Acres_Protected`, and `Miles_Protected`, were extracted from structured source fields when available. Quantitative values were also sometimes reported in project descriptions and could differ from the corresponding structured fields. Because project descriptions could have been written before restoration was implemented, these narrative values did not necessarily reflect final project outcomes. When structured values differed from those in project descriptions, the structured field values were retained, and the original descriptions were preserved without modification.

For sources compiled from reports or other unstructured project documentation, quantitative restoration metrics were extracted manually. Values explicitly reported in the available documentation were recorded in the corresponding database fields. When metrics were not clearly reported, the fields were left unassigned (NA).

The `Reported_Project_Cost` variable was populated using reported project cost values when available or calculated from multiple source variables where necessary. Because restoration projects can involve multiple funding streams, reported or calculated values did not necessarily represent the full cost of a project. Some sources reported only the funding contribution provided by the organization maintaining the dataset, while information about contributions from other organizations was optional or incomplete.

When a source dataset reported multiple funding streams for a project, the reported amounts were combined to calculate `Reported_Project_Cost`. Funding information was not combined across separate source datasets. The resulting value represented the reported or source-derived project cost available when the data were obtained and compiled; it was not independently validated as the total cost of the project.

When a source explicitly identified a value as expenditures, `Reported_Project_Cost` represented the amount expended by the time the data were obtained and compiled, rather than the total funding originally awarded. When the source did not specify whether a single reported value represented expenditures or awarded funding, the value was retained without further interpretation.

Additional project output metrics, including socioeconomic measures, were incorporated when relevant and available (e.g., `Socioeconomic_Jobs_Created`, `Socioeconomic_Avoided_Travel_Losses`, `Socioeconomic_Avoided_Damage_Losses`, and `Socioeconomic_Properties_Benefited`). Not all supplementary metrics reported in the source datasets were retained in the final database.

2.4.5 Spatial attribute standardization

The `County` variable was populated using the county information reported in source datasets when available. For projects without a reported county but with geographic coordinates, county assignments were derived using a spatial join in QGIS with a Florida county boundary shapefile[16] extending to the state waterline. The county containing the project coordinates was assigned based on the point-polygon overlap.

For projects reporting multiple counties, the project coordinates were used to assign a single primary county using the same spatial join procedure. The original list of reported counties was retained in the `County_Multiple` variable.

Projects that underwent this spatial assignment were flagged using a binary indicator (`County_GIS`).

Spatial joins were also used to assign each project to an Estuarine Drainage Area[18] (`EDA`) based on its geographic coordinates.

2.5 Supplemental data procedures

Supplemental data procedures were used to improve the completeness of selected standardized variables when information was systematically absent from a source dataset. Variables targeted for supplementation included project location, project status, completion year, restoration metrics, and funding information. Following source dataset standardization, datasets were reviewed to identify these gaps. Supplemental information was obtained from associated source documentation, such as annual reports, or directly from data providers when reliable information was available. These procedures addressed variables systematically absent from a source dataset rather than isolated missing values within otherwise populated variables. Supplemental information was incorporated only when it could be reliably linked to the corresponding project and source dataset.

For the National Fish and Wildlife Foundation (NFWF) Coastal Resilience Dashboard dataset (NFWF_NCRFandECRF), information from the NFWF Coastal Resilience Online Data Platform[19], annual conservation investment reports[20,21], grant award summaries[22–24], and project-specific documentation[25] was used to populate `Funding_Authority_Amount` and `Reported_Project_Cost`, which were not available in the downloaded dataset. Information obtained through a direct request to NFWF was used to populate `Completion_Year`, `Expected_Completion_Year`, `Latitude` and `Longitude`.

For the NFWF Gulf Environmental Benefit Fund dataset (NFWF_GEBF), coordinates (`Latitude` and `Longitude`) were obtained through a direct request to NFWF.

For the Florida Fish and Wildlife Conservation Commission (FWC) Artificial Reef Program dataset (FWC_REEFS), the original dataset did not include comparable quantitative restoration metrics, such as `Acres_Restored` or `Miles_Restored`. Restoration footprint calculations provided by the FWC Artificial Reef Program team through a direct request were used to populate `Acres_Restored`. FWC also provided calculated project funding information used to populate `Main_Funding_Source`, `Funding_Program`, and `Reported_Project_Cost`. Further details of these calculations are provided in the associated report[7].

2.6 Deduplication and record merging

A three-stage deduplication procedure was developed to identify and reconcile duplicate records within and across source datasets while retaining unique project information. At each stage, suspected duplicates were reviewed to determine whether records represented the same restoration project. Confirmed duplicates were either merged into a composite record or removed when one entry was fully redundant. Merged records were assigned a `Merged_Record_ID` linking the constituent records, and redundant records were flagged using a binary indicator.

When duplicate records contained conflicting values, the most specific entry was retained for each variable where the available information supported that decision. For example, when the `Habitat_Type` for an oyster restoration project was recorded as both "oysters" and "bivalves," the more specific classification, "oysters," was retained. Records reviewed during the first stage were excluded from subsequent stages to avoid repeated review.

Stage 1: The first stage targeted duplicate records with identical or nearly identical project information, including records resulting from human error or redundant reporting across source datasets. A duplication key was generated using `Project_Name`, `Project_Description`, `Latitude` and `Longitude`. This key was used to identify potential duplicates across source datasets, which were then reviewed manually.

Stage 2: The second stage targeted records that potentially represented the same project but differed in metadata, such as project coordinates or descriptions. A less restrictive duplication key based only on `Project_Name` was used to identify additional potential duplicates across source datasets. Suspected matches were reviewed manually using the same approach as in the first stage.

Stage 3: The third stage involved expert review of selected records associated with complex, multi-year, or multi-component restoration projects. These records were identified through feedback from Florida-based habitat restoration practitioners and the expertise of working group

members. This review addressed differences in reporting resolution across source datasets, particularly cases where one source reported individual project components while another reported the project as a whole.

Records were removed or merged only when the available source information supported an informed determination that they represented the same project. When source information was too limited or incomplete to establish duplication confidently, records were retained to avoid inadvertently excluding potentially distinct projects. Decisions regarding duplicate identification, record removal, and record merging were documented and justified.

2.7 Post-processing quality control

Following deduplication and record merging, additional processing was conducted to finalize standardized and project-level variables and apply exclusion criteria that had been deferred during initial screening. This included reviewing project status across merged records, incorporating additional information provided by data providers or identified by the working group, and recalculating derived variables when necessary to reflect the final project records. For example, records with a status of "Terminated" were retained during initial deduplication because a recently terminated record in one source could correspond to an active record in another source that had not been updated as recently. Additional records were excluded when subsequent source review or working group consensus indicated that they did not meet the database criteria. Derived and categorical values, including project size classifications, were recalculated when necessary to reflect updated quantitative values following record merging.

Project-level variables were derived from the final records associated with each project. These included restoration acreage and mileage, size categories, the number of habitat types, and completion year.

A source-level data quality review was also conducted for standardized records retained from each original dataset. Records were assessed for completeness of key project variables and whether the available information was sufficient to support reliable representation in the database. Consideration was also given to the ability to verify, maintain, and update project-specific information over time. Two source subsets were identified for further review.

The Florida Department of Environmental Protection Environmental Resource Permitting dataset (FDEP) was flagged because of limited completeness in key standardized variables. Following review, these records were excluded because the available source variables did not sufficiently align with the standardized database structure and overall objectives.

The Science for Nature and People Partnership–National Center for Ecological Analysis and Synthesis working group dataset (SNAPP) was flagged because most of its records had been

identified as duplicates of records from primary sources, while the remaining records contained limited information. Given that the data source was a static, secondary source used primarily to identify potentially missing datasets, the remaining records were ultimately excluded.

Finally, the geographic boundaries defined in section 2.3.2 were applied to the post-processed database using GIS-based spatial screening. This final screening accounted for changes to project information introduced during record merging. Projects with coordinates outside the defined geographic boundaries were excluded from the database v1.0.0.

3. Intended use and limitations

The Florida Estuarine Restoration Database was developed to support analyses of restoration activities across projects, organizations, and geographic areas in Florida estuaries and their contributing watersheds. It was not intended to serve as a comprehensive record of all information available for individual projects. The database includes information selected from source datasets and supplemental documentation available at the time of data acquisition; not all variables reported by the original sources were retained.

Database completeness and accuracy depend on the information provided by source organizations, their reporting practices, and the frequency with which their datasets are updated. Missing values indicate that information was not available or accessible during data compilation and do not necessarily indicate that the information was never collected. Although extensive efforts were made to identify and reconcile duplicate records, some duplicates may remain. Project statuses and other attributes were not independently verified, and the database reflects the information available from the sources at the time of acquisition. Users seeking current or project-specific information should consult the primary source using the project URL or contact information provided in the database, when available.

4. Data maintenance and version control

The Florida Estuarine Restoration Database will be updated annually to incorporate newly available restoration project information, corrections, and updates to existing records. Each release will be assigned a version number to support reproducibility and allow users to identify the database version used in an analysis.

Changes between releases will be documented in the CHANGELOG.md file maintained in the project's GitHub repository. The changelog will include the release date, data sources added or updated, changes to project records, and any substantial changes to processing methods or database structure.

New records and updates to existing records will follow the standardization, screening, and quality control procedures described in this protocol to maintain consistency with the established database structure and inclusion criteria. Any substantial changes to the database structure or processing methods will be documented in subsequent protocol versions.

5. References

1. R Core Team. R: A Language and Environment for Statistical Computing, version 4.4.1. https://www.R-project.org/ (2024).
2. QGIS Development Team. QGIS Geographic Information System, version 3.34.1. https://qgis.org (2023).
3. DIVER. Web Application: Data Integration Visualization Exploration and Reporting Application, National Oceanic and Atmospheric Administration. (accessed 4 February 2026) https://www.diver.orr.noaa.gov/web/guest/diver-explorer?siteid=9&sqid=643&subtitle=DWH%20Restoration%20Projects.
4. Ducks Unlimited, Gulf of Mexico Alliance & The Trust for Public Land. Deepwater Horizon Project Tracker. Gulf of Mexico Alliance (accessed 11 December 2025) https://dwhprojecttracker.org/download-project-information.
5. Florida Department of Environmental Protection (FDEP). Environmental Resource Permitting Coordination, Assistance, Portals. (accessed 19 May 2025) https://floridadep.gov/water/submerged-lands-environmental-resources-coordination/content/environmental-resource-permitting.
6. Florida Fish and Wildlife Conservation Commission (FWC). Marine and Estuarine Habitats Project Map. Florida Fish and Wildlife Conservation Commission, Aquatic Habitat Conservation and Restoration Section (accessed 1 March 2026) https://myfwc.maps.arcgis.com/apps/webappviewer/index.html?id=ddaedd01801147a1ae40ad6e1173b3be.
7. Rodriguez, Rae & Mille, Keith. Florida Artificial Reef Estuarine Deployment Footprints with Cost: Background and supporting documentation. Florida Fish and Wildlife Conservation Commission Artificial Reef Program, Division of Marine Fisheries Management (accessed 23 June 2026) https://doi.org/https://knb.ecoinformatics.org/view/doi%3A10.5063%2FF1765CTW (2026).

8. Florida Fish and Wildlife Conservation Commission (FWC). Artificial Reefs in Florida. Florida Fish and Wildlife Conservation Commission, Artificial Reef Program https://geodata.myfwc.com/datasets/artificial-reefs-in-florida/about (2024).
9. National Aquatic Connectivity Collaborative (NACC). Barrier Inventory & Prioritization Tool. National Aquatic Connectivity Collaborative (accessed 4 February 2026) https://tool.aquaticbarriers.org/explore/ (2026).
10. U.S. Environmental Protection Agency (EPA). National Estuary Program Dashboard. U.S. Environmental Protection Agency (accessed 6 January 2026) https://www.epa.gov/nep/nepmap (2026).
11. National Fish and Wildlife Foundation (NFWF). Coastal Resilience Dashboard. National Fish and Wildlife Foundation https://resiliencedashboard.nfwf.org (2025).
12. National Fish and Wildlife Foundation (NFWF). Gulf Environmental Benefit Fund: Projects. (accessed 1 July 2026) https://www.nfwf.org/gulf-environmental-benefit-fund/projects.
13. National Oceanic and Atmospheric Administration (NOAA). Restoration Atlas. NOAA Fisheries (accessed 16 March 2026) https://www.habitat.noaa.gov/apps/restoration-atlas/ (2025).
14. National Center for Ecological Analysis and Synthesis. SNAPP: Aligning coastal restoration with ecological and societal needs. National Center for Ecological Analysis and Synthesis (accessed 21 April 2025) https://www.nceas.ucsb.edu/workinggroups/snapp-aligning-coastal-restoration-ecological-and-societal-needs.
15. Florida Department of Environmental Protection (FDEP). Numeric Nutrient Criteria (NNC) - Estuary Nutrient Regions. Florida Department of Environmental Protection Open Data Portal https://geodata.dep.state.fl.us/datasets/FDEP::numeric-nutrient-criteria-nnc-estuary-nutrient-regions/about (2014).

16. Florida Fish and Wildlife Conservation Commission-Fish and Wildlife Research Institute (FWC-FWRI). Florida County Boundaries 1:24000. (accessed 1 September 2026) https://www.arcgis.com/home/item.html?id=404a8909bd6f42259c0e83d6d31c3f12#overview (2025).
17. Florida Department of Environmental Protection (FDEP). HUC10 Model Coverage. Florida Statewide Flood Model Inventory Dashboard https://www.arcgis.com/home/item.html?id=3530aaa50d8f45f8812eb14f925e0ca1#overview (2025).
18. Dale, L. L. *et al.* Estuarine drainage area boundaries for the conterminous United States. U.S. Geological Survey data release https://doi.org/10.5066/P9LPN3YY (2022).
19. National Fish and Wildlife Foundation (NFWF). Coastal Resilience Open Data Platform. https://resiliencedata.nfwf.org/organization/?controller=organization&action=index.
20. National Fish and Wildlife Foundation (NFWF). 2020 Conservation Investments. (National Fish and Wildlife Foundation, 2020). https://www.nfwf.org/sites/default/files/2021-03/2020-conservation-investments.pdf
21. National Fish and Wildlife Foundation (NFWF). 2021 Conservation Investments. (National Fish and Wildlife Foundation, 2021). https://www.nfwf.org/sites/default/files/2022-03/NFWFci20220304.pdf
22. National Fish and Wildlife Foundation (NFWF). National Coastal Resilience Fund: November 2022 Grant Slate. (National Fish and Wildlife Foundation, 2022). https://www.nfwf.org/sites/default/files/2022-12/NFWF-NCRF-20221129-Nov-GS.pdf
23. National Fish and Wildlife Foundation (NFWF). National Coastal Resilience Fund: November 2023 Grant Slate. (National Fish and Wildlife Foundation, 2023). https://www.nfwf.org/sites/default/files/2023-11/NFWF-NCRF-20231114-GS.pdf
24. National Fish and Wildlife Foundation (NFWF). National Coastal Resilience Fund: October 2023 Grant Slate. (National Fish and Wildlife Foundation, 2023). https://www.nfwf.org/sites/default/files/2023-10/NFWF-NCRF-20231002-GS-Oct.pdf
25. National Fish and Wildlife Foundation (NFWF). Pensacola Bay Living Shoreline Project Standard Quality Assurance Plan - V0.2. (National Fish and Wildlife Foundation, 2025).

https://resiliencedata.nfwf.org/dataset/9f314e7f-cc80-49a1-a0e0-ff4269cb7e90/resource/fb000e50-9b80-49fc-9e4f-d948829162bc/download/pbls-qapp-v0.2-20250402-.pdf

6. Appendix

Table A1. Data harmonization and standardization crosswalk for the database. The table documents the relationship between original source variables and standardized database variables, including data transformations, inferred values, calculation procedures, assumptions, known limitations, and quality-control notes used during database development. The table is not exhaustive and focuses on variables associated with these procedures; routine variables with clear, direct interpretations are generally omitted (e.g., latitude, longitude, project name). Definitions of original variables can be found in the original data source metadata on the GEI Github repository, if available.

Standardized Variable	Data Source Code	Original Variable(s)	Notes
Acres_Restored	DWH	`Land Restoration Acreage` and `Aquatic Restoration Acreage`	The standardized variable is calculated from the sum of the original variables `Land Restoration Acreage` and `Aquatic Restoration Acreage`.
Acres_Restored	FWC_REEFS	SqFt_Restored	The standardized variable was calculated from the original variable divided by 43,560.
Acres_Restored	NACC	WaterbodyAcres	The original variable represents the 'area in acres of waterbody associated with the given dam'.
Acres_Restored	NFWF_NCRFandECRF	`Acres Restored`	The value associated with the Original Variable (housed within the Original Variable 'Measure') is based on project status: (1) for active projects, the value represents the projected value; and (2) for completed projects, the value represents the actual value.
Acres_Protected	DWH	`Acres Acquired`	The original variable represents the 'number of acres gained in acquisition project'.
Completion_Year	NACC	YearRemoved	The original variable represents the 'year that barrier was removed or mitigated, if available'.

Completion_Year	NEP	Federal_Fiscal_Year	Informed by NEP staff that the federal fiscal year the project was entered into the database represents the completion year of the project. Therefore, all projects were additionally labeled as 'Complete' for the 'Project_Status' variable.
Completion_Year	NOAA	`Fiscal Year`	Informed by NOAA staff that the fiscal year represents the year the acreage for a given habitat of a given project was reported "complete". This allows for a project to have multiple fiscal years representing the acreage complete across different habitats as the project progresses. Following NOAA staff recommendation, the complete year of a given project was represented as the latest fiscal year reported on a given project.
Completion_Year	NFWF_GEBF	Completion_Year	If the value (date) for the original variable had already occurred, it was added to the standardized variable. If the value had not yet occurred, it was added to the standardized variable 'Exp_Completion_Year'.
Completion_Year	NFWF_NCRF andECRF	Project_End_Date	NFWF staff informed us that the original variable represents 'the date the grantee expects the work to be done; grantees must request an amendment if it changes significantly.' Therefore, if the value (date) for the original variable had already occurred, it was added to the standardized variable. If the value had not yet occurred, it was added to the standardized variable 'Exp_Completion_Year'.
Exp_Completion_Year	DWH	`Anticipated Project Completion Year`	
Exp_Completion_Year	NFWF_GEBF	Completion_Year	If the value (date) for the original variable had not yet occurred, it was added to the standardized variable. If the value had already occurred, it was added to the standardized variable 'Completion_Year'.
Exp_Completion_Year	NFWF_NCRF andECRF	Project_End_Date	NFWF staff informed us that the original variable represents 'the date the grantee expects the work to be done; grantees must request an amendment if it changes significantly.' Therefore, if the value (date) for the original variable had not yet occurred, it was added to the standardized variable. If the value had already occurred, it was added to the standardized variable 'Completion_Year'.

Habitat_Type	DWH	`Primary Targeted Resource`	The original variable represents ‘'a description of the resources or habitats benefitted by the project.’ For projects where there was no value, the value for the standard variable of a given project was inferred if clearly defined in the project description. If there was any ambiguity, the standard variable value was left as NA. **Caveat:** This could result in misidentification of the value of the standard variable due to human error.
Habitat_Type	FWC_REEFS	NA	Given the source of the data, the value of the standardized variable was set to 'hard bottom'.
Habitat_Type	NACC	NA	Given the source of the data, the value of the standardized variable was set to "tidal tributary/creek/river", as all selected projects occurred in creeks/streams.
Habitat_Type	NFWF_NCRF andECRF	sub_category	
Lead_Implementer	NFWF_NCRF andECRF	grantee	
Lead_Implementer	NOAA	`Funding Recipients`	
Main_Funding_Source	DWH	`DWH Funder`	The standardized variable was assigned a value based on the largest funding contribution. Monetary funds were compared across original variables `DWH Funder`, `Funder Name 2`, `Funder Name 3` etc. If a funder other than `DWH Funder` had the largest contribution, the value of `DWH Funder` was added to the list of original variable `Other Partners`.
Main_Funding_Source	FWC_REEFS	`Funding Source - Entity`	The original variable represents ‘the name of the entity that served as a funding source for each artificial reef deployment.’ Although categorical funding information was provided (e.g., `Local Govt Funding (byDeployID)`, `State Funding (byDeployID) `, `Federal Funding (byDeployID) `,`Other Funding`), the dataset did not provide the names of other partners for a given project and, as a result, prevented calculations to distinguish the primary funding source

			based on the proportion of total project funding (e.g., >50% of total reported project cost). Therefore, the value from the original variable `Funding Source - Entity` was assigned the value of the standardized variable Main_Funding_Source by default. **Caveat:** There is one instance where the Main_Funding_Source does not represent the entity contributing the largest funding proportion (>50%) of the total reported project cost.
Main_Funding_Source	NOAA	NA	Due to confidentiality and data sharing limitations of the original data source, funding information was only provided through the original variables `NOAA Contributions (project total)` and `Sum of Other Contributions` and prevented comparison of other individual funding contributions. Therefore, NOAA was assigned as the Main_Funding_Source for all projects from this data source. **Caveat:** There may be projects where the NOAA Office of Habitat Conservation is not the primary funding contributor.
Miles_Restored	NOAA	`Actual Stream Miles Opened`	
Miles_Restored	NFWF_NCRF andECRF	`Miles Restored & Created`	The value associated with the Original Variable (housed within the Original Variable 'Measure') is based on project status: (1) for active projects, the value represents the projected value; and (2) for completed projects, the value represents the actual value.
Original_Project_ID	DIVER	Project_ID	
Original_Project_ID	DWH	OBJECTID	
Original_Project_ID	FDEP	`Application #`	
Original_Project_ID	FWC_REEFS	DeployID	Based on guidance from FWC staff, it was determined that each DeployID should be considered a distinct restoration project, rather than linking records

			by grant. This approach was selected because it is very common for a single grant to fund numerous restoration projects spanning large geographic areas and multiple years. Grouping records at the grant level for this specific dataset would therefore reduce the spatial and temporal resolution of the database. Given the level of detail available for project location, area, and funding, DeployID was determined to be the most appropriate unit for defining individual restoration projects.
Original_Project_ID	NACC	SARPID	
Original_Project_ID	NEP	View_Detail; ID_Number	The original variable 'View_Detail' represents the Original_Project_ID for dataset 'raw_NEP', and the original variable 'ID_Number' represents the Original_Project_ID for dataset 'raw_NEP_pre2006'.
Original_Project_ID	NFWF_NCRF andECRF	ez_grant_id	
Original_Project_ID	NOAA	Project Number	
Original_Project_ID	SNAPP	Master_Project_ID	
Restoration_Technique	DWH	NA	The value for the standard variable for a given project was inferred if clearly defined in the project description. If there was any ambiguity, the standard variable value was left as NA. **Caveat**: This could result in misidentification of the value of the standard variable due to human error.
Restoration_Technique	FWC_REEFS	NA	Given the source of the data, the value of the standardized variable was set to 'reef construction'.
Restoration_Technique	NACC	NA	Given the source of the data, the value of the standardized variable was set to `hydrologic restoration`, as all selected projects were dam removal.
Reported_Project_Cost	DWH	`Dollar Amount` and `Total Leveraged Funds`	The standardized variable is calculated from the sum of the original variables `Dollar Amount` and `Total Leveraged Funds`, where `Total Leveraged Funds` represents the 'total leveraged (non-DWHPT) dollars used to fund the project

			summed across individual leveraged sources'.
Reported_Project_Cost	NOAA	`NOAA Contributions (project total)` and `Sum of Other Contributions`	The standardized variable is calculated from the sum of the original variables `NOAA Contributions (project total)` and `Sum of Other Contributions`. **Caveat:** Information on additional funding contributions from other sources may be provided through the NOAA project reporting process, but it is not required. Therefore, `Sum of Other Contributions` may not include all additional funding associated with a given project, and the resulting standardized variable Reported Project Cost may not represent the true total project cost.
Start_Year	NFWF_NCRFandECRF	year_awarded	
Socioeconomic_Avoided_Travel_Losses	NFWF_NCRFandECRF	`Avoided Travel Time Losses (Thousands $)`	The standardized variable is calculated from the sum of the original variable for auto and rail avoided travel time losses by the project. The values were converted thousands of US dollars to standard US dollars.
Socioeconomic_Avoided_Damage_Losses	NFWF_NCRFandECRF	`Avoided Property Damage & Reduced Repairs (Millions $)`	The standardized variable is calculated from the sum of the original variable for commercial and residential avoided property damages and repaired by the project. The values were converted millions of US dollars to standard US dollars.
Socioeconomic_Jobs_Created	NFWF_NCRFandECRF	`Jobs Created`	The standardized variable is calculated from the sum of the original variable for direct, indirect and induced jobs created by the project.
Socioeconomic_Properties_Benefitted	NFWF_NCRFandECRF	`Total # of Properties Benefitted from Enhanced Protection`	The standardized variable is calculated from the sum of the original variable for commercial, residential, public, and critical infrastructure that benefitted from the project.